

Excel Formulas — Detailed Lecture Notes

JKSSB Computer Awareness

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25.1 What a formula is

A **formula** is an instruction that tells Microsoft Excel how to calculate the value of a cell. Instead of holding a fixed number, the cell holds a rule, and Excel works out the result of that rule from the other cells, from constants, or from functions the rule refers to. The formula is typed into the formula bar; the result is shown in the cell. When any cell that the formula uses changes, Excel recalculates and shows the new result automatically.

Definition (Formula). *An instruction, written in a cell or the formula bar, that calculates a value from operands such as cell references, constants, and functions. The cell stores the formula, while the sheet displays the result.*

This lesson covers the four building blocks of a formula: the leading =, the operators, the cell references, and the way references behave when a formula is copied. It closes with the standard error codes and the distinctions that examiners test most often.

25.2 The = rule and formula syntax

Every Excel formula must begin with the equal sign =. This is the single fact most often tested on this topic. Typing =A1+A2 into a cell is treated as a formula and calculated; typing A1+A2 without the leading = is treated as ordinary text and shown exactly as written.

The same = symbol also serves as a comparison operator that tests whether two values are equal. The position tells you which role it plays: at the very start of an entry it begins a formula, while inside a formula it compares two operands.

Definition (The = rule). *A formula must start with =. A cell entry that does not start with = is not a formula.*

A simple formula has three parts: the leading =, the operands, and the operators. In the formula =A1*B1+100, the operands are the references A1 and B1 and the constant 100, while the operators are * (multiply) and + (add).

25.3 Formula, function, and constant

It helps to separate three ideas that often appear together. A **constant** is a fixed value written directly into a formula, such as the number 100 or a piece of text in quotation marks. A **function** is a built-in calculation with a name, such as SUM or AVERAGE, which is called from inside a formula. A **formula** is the whole instruction, and it may combine constants, references, operators, and functions in one expression.

Definition (Function). A named, built-in calculation such as *SUM(A1:A10)* or *AVERAGE(B1:B5)* that is called from within a formula and returns a single value.

The distinction matters for exams because a question may ask what a formula contains rather than what it produces. The formula `=A1+A2` contains two relative references and one arithmetic operator. The formula `=SUM(A1:A10)` calls a function on a range. Both begin with `=`, and both return a value to the cell that holds them. The rest of this lesson concentrates on the references and operators that every formula uses.

25.4 Operators

An operator tells Excel what to do with the values on either side of it. Excel has four families of operators: arithmetic, comparison, text, and reference.

Arithmetic operators produce numbers.

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponent (raise to a power)
%	Percent

Two symbols are easy to get wrong. Multiplication uses the asterisk `*`, not the letter `x` or a dot; division uses the forward slash `/`. The caret `^` raises a number to a power, so `=2^3` is 8. The percent sign `%` divides by 100, so `=50%` is 0.5.

Comparison operators compare two values and return **TRUE** or **FALSE**: `=` (equal to), `>` (greater than), `<` (less than), `>=` (greater than or equal to), `<=` (less than or equal to), and `<>` (not equal to). The operator `<>` is the Excel way of writing “not equal”; it is not written `!=`.

A comparison gives a useful result even on its own: if A1 holds 50, the formula `=A1>100` shows **FALSE**. The operators that candidates confuse most are `=`, which tests equality when it appears inside a formula, and `<>`, which tests inequality. Note that the leading `=` that starts every formula and the `=` comparison operator are the same character used in two different roles.

Text operator. The ampersand `&` joins, or concatenates, two pieces of text into one. `=Total:&A1` shows the words **Total**: followed by the contents of cell A1.

Reference operators build references to groups of cells. The colon `:` is the range operator, so `A1:A10` means every cell from A1 down to A10. The comma `,` is the union operator, which joins two ranges into one list. A single space between two references is the intersection operator, which returns the cells common to both ranges.

The four families can be summarised in one place. Arithmetic operators (`+` `-` `*` `/` `^``%`) produce numbers. Comparison operators (`=` `>` `<` `>=` `<=` `<>`) produce **TRUE** or **FALSE**. The text operator `&` produces a joined piece of text. The reference operators `:`, `,`, and the space build ranges and lists of cells. Knowing which family an operator belongs to tells you what kind of result to expect.

25.5 Order of evaluation (precedence)

When a formula has several operators, Excel does not simply work from left to right. It follows a fixed order of precedence. Parentheses are evaluated first. Then the percent sign, then the

exponent, then multiplication and division, then addition and subtraction. After the arithmetic comes the concatenation operator &, and finally the comparison operators.

Because multiplication and division are evaluated before addition and subtraction, $=2+3*4$ is 14, not 20. The multiplication $3*4$ is done first to give 12, and then 2 is added. In the same way $=2+3*4$ differs from $=(2+3)*4$, which is 20. Parentheses are the reliable way to control the order of evaluation.

Example 25.5.1. *In $=10+2*5$ the multiplication is done first, so the result is 20. If the intention was to add first, the formula must be written $=(10+2)*5$, which gives 60. The exponent binds even tighter than multiplication, so $=2*3^2$ is 18, because 3^2 is evaluated before the multiplication by 2.*

25.6 Cell references

A **cell reference** names a cell by its column letter and row number. In the default **A1** reference style, **A1** is column A, row 1, and **B5** is column B, row 5. A reference lets a formula use the current value of another cell, so the formula stays correct when that value changes.

The crucial property of a reference is not what it points at now but how it behaves when the formula is *copied*. There are three kinds of reference: relative, absolute, and mixed.

25.7 Relative references

A reference such as **A1** is **relative**. Excel stores it, in effect, as an offset from the cell that contains the formula. When the formula is copied to another cell, the reference shifts by the same distance, so it keeps pointing at the cell that lies in the same relative position.

Example 25.7.1. *If $=A1$ is entered in cell B1 and then copied to B2, the formula becomes $=A2$. The reference moved down one row because the copied formula moved down one row. Copy the same formula from B1 to C1, and it becomes $=B1$: the reference moved one column to the right because the formula did.*

Relative references are the default and are what you want for a column of repeated calculations, such as adding each row's two numbers.

25.8 Absolute references

A reference such as **\$A\$1** is **absolute**. The dollar sign locks the part of the reference that follows it: the first \$ locks the column, and the second \$ locks the row. An absolute reference does not change when the formula is copied; it always points at the same cell.

Example 25.8.1. *If $=$A1 is entered in B1 and copied to B5, it remains $=$A1 . Neither the column nor the row moved. A fixed value such as a tax rate or a constant kept in one cell should be referenced absolutely, so that copying the formula does not disturb it.*

25.9 Mixed references

A **mixed** reference locks only one part. In **\$A1** the column is locked and the row is relative; copying the formula to another row changes the row number but never the column. In **A\$1** the row is locked

and the column is relative; copying the formula to another column changes the column letter but never the row.

A useful way to remember this is that the \$ locks whatever immediately follows it. A \$ before a letter locks the column, and a \$ before a number locks the row.

25.10 Copying and filling formulas

The behaviour of a reference under copying is the heart of the topic. When a formula is copied, only the unlocked parts of each reference shift; the locked parts stay put. The table below compares the four reference types when a formula is copied one column right and one row down, from cell B1 to cell C2.

Reference	What is locked	After copying B1 to C2
A1	nothing	B2
\$A\$1	column and row	\$A\$1
\$A1	column	\$A2
A\$1	row	B\$1

Reading the table: the relative reference A1 moved on both axes; the absolute reference \$A\$1 moved on neither; \$A1 kept its column but followed the row; and A\$1 kept its row but followed the column.

A second, very common direction is copying a formula straight down a column. If =A1*B1 sits in C1 and is copied down to C2, the formula becomes =A2*B2: both relative references followed the copy down one row. If the value in B1 must stay fixed, the formula should instead read =A1*\$B\$1, so that only A1 follows the copy while \$B\$1 stays anchored.

The same four references behave as follows when copied one column to the right, and when copied one row down.

Reference	Copied right (B1→C1)	Copied down (B1→B2)
A1	B1	A2
\$A\$1	\$A\$1	\$A\$1
\$A1	\$A1	\$A2
A\$1	B\$1	A\$1

Reading either column, the rule is the same: a reference moves only along the axis that is not locked. The relative reference moves on both axes, the absolute reference on neither, and each mixed reference on exactly one.

25.11 Worked examples

Example 25.11.1. A price list has prices in column A and a fixed tax rate stored in cell \$B\$1. The formula in cell C2 is =A2*\$B\$1. When it is copied down, C3 becomes =A3*\$B\$1 and C4 becomes =A4*\$B\$1: the price reference moves down with each row, while the rate reference stays locked on \$B\$1. If the formula had been written =A2*B1, copying it down would have moved the rate to B2 and then B3, giving wrong results.

Example 25.11.2. In cell B1 the formula is `=A1`. It is copied to C2. The column lock holds A, and the relative row follows the copy down one row, so C2 contains `=A2`. If instead the original formula were `=A$1`, copying from B1 to C2 would keep the row 1 and move the column one step from A to B, giving `=B$1`.

Example 25.11.3. A sheet lists marks in column B. To flag each student as passing, the formula `=B2>=40` is entered in C2 and copied down. It returns **TRUE** for a mark of 40 or more and **FALSE** otherwise. Because B2 is relative, the reference follows the copy down the column, so C3 tests B3 and C4 tests B4. The comparison operator produces the **TRUE/FALSE** result, not a number.

Example 25.11.4. A score sheet holds marks in B2:B11 and the class total in cell `$B$12`. To show each student's share as a percentage, the formula in C2 is `=B2/$B$12*100`. The relative B2 follows the copy down the column, while `$B$12` stays anchored on the total. The operators here are division and multiplication, evaluated from left to right at the same precedence, and the result is a plain number. If the total were written as a relative reference, copying the formula down would corrupt the denominator.

25.12 Common formula errors

When a formula cannot be evaluated, Excel shows an error value in the cell. The codes tested in exams are:

Error	Meaning
#DIV/0!	Division by zero, or by an empty cell.
#VALUE!	The wrong type of operand, such as text where a number is needed.
#REF!	A referenced cell was deleted, so the reference is no longer valid.
#NAME?	An unknown function, or a misspelled name or text entry.
#N/A	The value is not available, as when a lookup finds no match.
#NULL!	The two ranges given do not intersect.

Remember the two that examiners use most: **#DIV/0!** for division by zero and **#REF!** for a reference that has been broken by deleting a cell. A related warning is the **circular reference**, which appears when a formula refers, directly or through a chain of other formulas, back to its own cell, so Excel cannot work out a value. Deleting the referenced row or column is the usual cause of **#REF!**, while supplying the wrong type of operand, such as text where a number is expected, produces **#VALUE!**.

25.13 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A formula must start with `=` (TRAP-30, PYQ-38). The options `+`, `%`, and `@` are distractors.
2. `$` before a letter locks the column; `$` before a number locks the row. Do not swap these.
3. **Relative** references change on copy; **absolute** references do not. A **mixed** reference changes on one axis only.
4. **#REF!** means a broken reference; **#DIV/0!** means division by zero. Do not swap the error codes.

5. <> is the Excel operator for “not equal”; multiplication is * and division is /.
6. A **function** such as SUM is called from inside a formula, and a **constant** is a fixed value written directly; both are operands, not operators.

Example 25.13.1. *An item asks what =\$A\$1 becomes when copied from B1 to B5. Because both the column and the row are locked by the two dollar signs, the reference does not move: the answer is =\$A\$1. If the question instead gives =A1, the column is locked but the row is relative, so the answer is =A5. The dollar signs, not the row numbers, decide the answer.*

Exercises

1. Which character must every Excel formula begin with, and what happens if it is omitted?
2. List the arithmetic operators for addition, subtraction, multiplication, division, and exponent.
3. Explain the difference between a relative reference A1 and an absolute reference \$A\$1.
4. In the mixed reference \$A1, which part is locked and which part changes when the formula is copied?
5. The formula =A1 is in cell B1. What does B2 contain after the formula is copied to it? What would B2 contain if the formula were =\$A\$1?
6. State the meaning of the errors #DIV/0! and #REF!.
7. What is the result of the formula =2+3*4, and why?
8. Name the Excel operator that joins two pieces of text, and the operator that means “not equal to”.

Solutions

1. A formula must begin with =. Without it, the entry is stored and shown as ordinary text rather than calculated.
2. Addition +, subtraction −, multiplication *, division /, and exponent ^.
3. A1 is relative: on copying, both its column and its row shift by the same offset as the formula. \$A\$1 is absolute: the dollar signs lock the column and the row, so it does not change on copying.
4. In \$A1 the column A is locked; the row is relative, so the row number changes when the formula is copied to a different row.
5. Copying =A1 from B1 to B2 gives =A2, because the relative reference follows the copy down one row. Copying =\$A\$1 to B2 leaves it as =\$A\$1, because both parts are locked.
6. #DIV/0! means division by zero or by an empty cell; #REF! means a reference is invalid because a referenced cell was deleted.
7. The result is 14. Multiplication is evaluated before addition, so 3*4 is 12, and 2+12 is 14.
8. The ampersand & joins (concatenates) text, and <> means “not equal to”.